



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 01 – “Resistance Zero”

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*“Any sufficiently advanced technology
is indistinguishable from magic.”*

— Arthur C. Clarke

“Profiles of the Future: An Inquiry into the Limits of the Possible”

A levitating magnet



Figure 1: A magnet levitating on a superconductor around 77 K (Paris-Saclay).

Levitation is ubiquitous



Figure 2: Left: a frog levitating in a magnetic field (Berry & Geim); Middle: light styrofoam particles trapped in a standing sound waves (Wikipedia); Right: a person floating on air pressure (iFLY Toronto).

Suspension !?



Figure 3: A superconductor hanging below a magnet (Paris-Saclay).

Back in 1911 ...

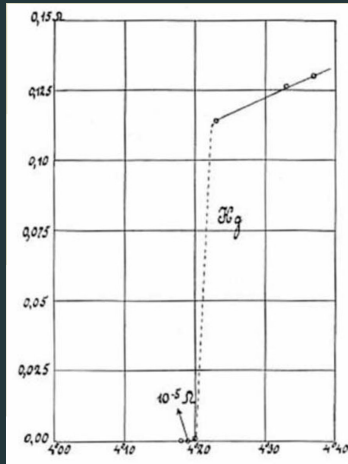


Figure 4: Electrical resistance (ohms) of mercury as a function of temperature (in kelvin) (Onnes & Keesom)

Superconductivity

Simply put:

- $\rho = 0$ (lossless electrical transport),
- $B = 0$ (magnetic levitation).

Interesting physics:

- **Macroscopic** quantum phenomena
- Synthesis of QM, Stat. Phys., & EM.

Gigantic impact:

- MRI, voltage standard
- LHC, Fusion, etc.
- Quantum computer

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Why study Superconductivity?

Societal impact: Health

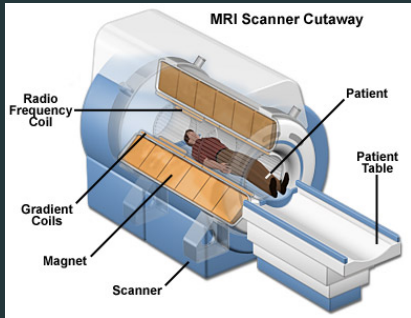


Figure 5: MRI scanners use SC magnets (nationalmaglab) capable of generating magnetic fields about **one hundred thousand times stronger** than the Earth's surface field.



Figure 6: Brain activity can be monitored with MEG systems based on SQUID technology (Elekta), capable of detecting magnetic fields a billion times weaker than the Earth's field.

Societal impact: Energy

At ITER, plasma confinement relies on a solenoid SC coil measuring 18 m in height, 4.3 m in diameter, and weighing 1,000 tonnes.

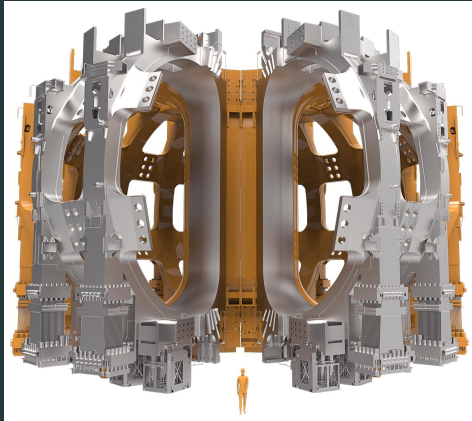


Figure 7: A toroidal field system of ITER

Societal impact: Science

The LHC employs around **10,000** SC magnets, with a combined weight of 470 tonnes, to steer the particle beams.

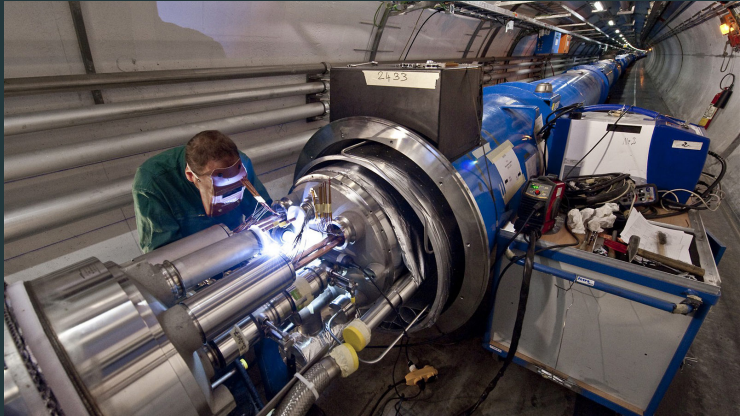


Figure 8: Superconducting electromagnets (CERN)

Societal impact: Technologies

In 2019, a UC Santa Barbara team in partnership with Google reported achieving quantum supremacy using a chip with 53 SC qubits.

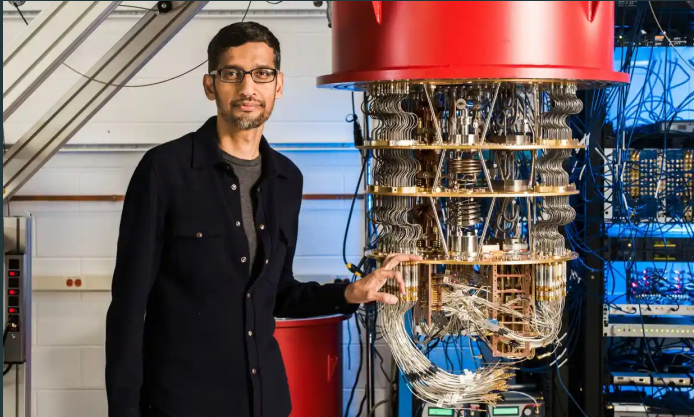


Figure 9: Google's CEO and quantum computer (Google)

Why study superconductivity

- Outstanding fundamental questions:
 - *How are two electrons bound together?*
 - We have a solid understanding (to be covered in this module),
 - yet much remains unknown! (active research)
- Rich interplay between theory and experiment

Discovery

Where it all began

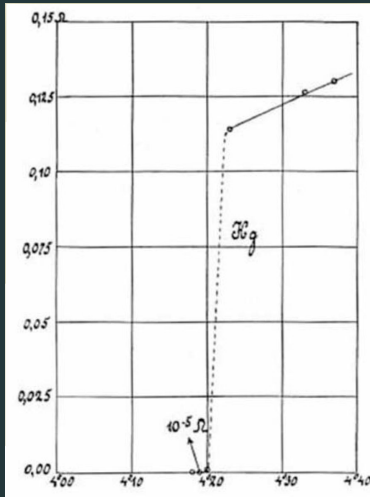


Figure 10: The $R(T)$ of mercury

- What would you have expected? Are you surprised?
- Why was mercury chosen?
- Is the resistance truly zero?
- What would you investigate next after this discovery?

Not as simple as it looks

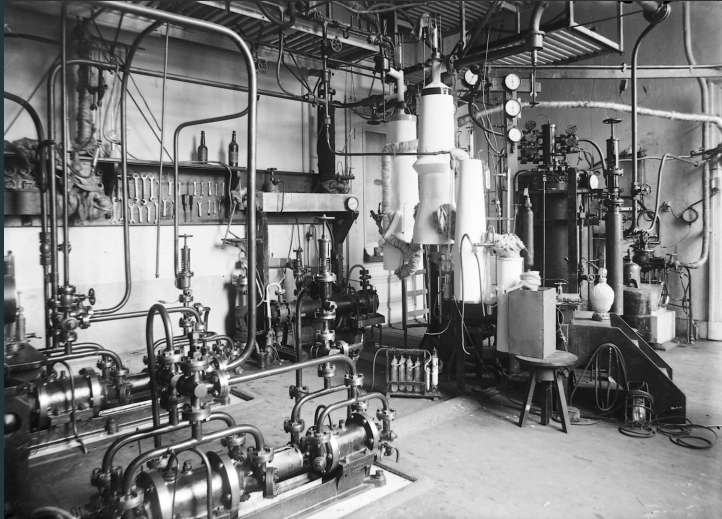


Figure 11: The K. Onnes laboratory @ Leiden, Netherlands.

Periodic table

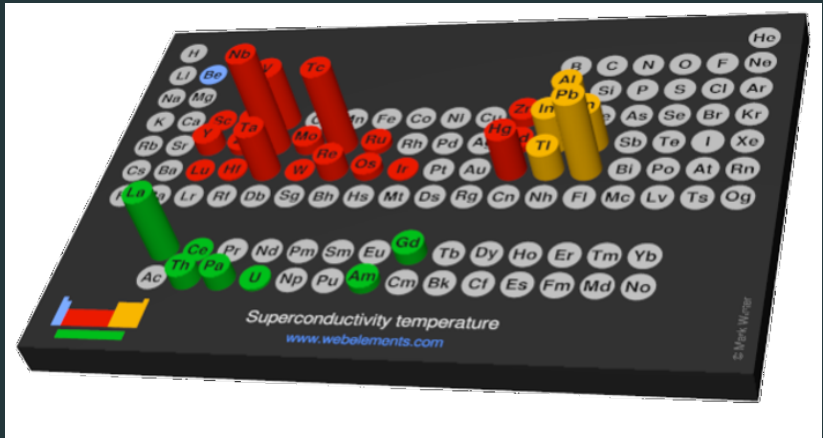


Figure 12: Bar height represents the superconducting transition temperature of each element.

Timeline

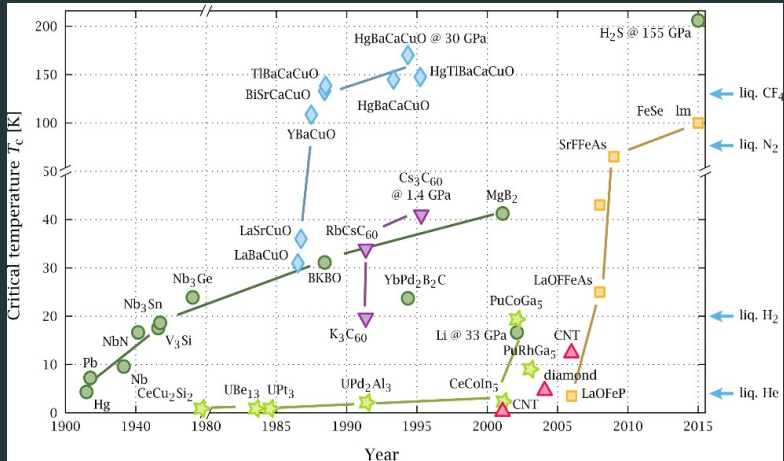


Figure 13: Timeline of superconductors, with colours indicating different material classes or origins (Wikipedia).

Room temperature superconductivity!

Reported in 2020 by the Fenster group, Univ. Rochester, USA:



Figure 14: A superconductor mounted inside a diamond anvil pressure cell.

Just a bit of pressure (267 GPa), equivalent to the pressure at outer core of Earth. However, the *Paper Retracted* !

Reference

The primary text for this module is **LECTURE NOTES**. In addition, you are expected to consult relevant sections of the following literature:

1. AC Rose-Innes & EH Rhoderick, *Intro. to Superconductivity*,
2. JE Annett, *Superconductivity, Superfluids and Condensate*,
3. JR Waldram, *Superconductivity in metals and cuprates*,
4. W Buckel & R Kleiner, *Superconductivity*,